

MAT1002: Calculus II L02

Quiz 1 — 2025-2026, Term 2

For Tutors Only — Not For Distribution

Wednesday, 28 Jan 2026

1. (a) (6 points) Show that $\{(-1)^n\}_{n=1}^{\infty}$ diverges.

(b) (4 points) Find

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n}.$$

Solution:

(a) *Proof.* Suppose not. Then $\lim_{n \rightarrow \infty} (-1)^n = L$ for some $L \in \mathbb{R}$. Let $\epsilon = 1$. By definition, there exists N such that for all $n > N$, we have $|(-1)^n - L| < 1$. In particular, when n is even, we have $|L - 1| < 1$, i.e. $0 < L < 2$; if n is odd, we instead have $|L + 1| < 1$, i.e. $-2 < L < 0$. Such an L does not exist, which is a contradiction. Hence, $\{(-1)^n\}$ diverges. $\square$

(b)

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0,$$

where to get the 2nd equality, we have used L'Hôpital's rule.

2. Justify whether the following series converges. If so, calculate the limit.

(a) (2 points)

$$\sum_{n=1}^{\infty} \frac{e^n - 2}{4e^n + 5}.$$

(b) (2 points)

$$\sum_{n=1}^{\infty} \sin \frac{n\pi}{2}.$$

(c) (3 points)

$$\sum_{n=1}^{\infty} \frac{1}{n}.$$

(d) (3 points)

$$\sum_{n=3}^{\infty} \sin \frac{1}{n}.$$

Solution:

(a) The series diverges since

$$\lim_{n \rightarrow \infty} \frac{e^n - 2}{4e^n + 5} = \frac{1}{4} \neq 0.$$

(b) The series diverges since

$$\lim_{n \rightarrow \infty} \sin \frac{n\pi}{2}$$

does not exist.

(c) Consider $f(x) = 1/x$, which is positive and decreasing on $[1, \infty)$. Since

$$\int_1^{\infty} \frac{1}{x} dx$$

diverges, and by the integral test, we know that the harmonic series diverges.

(d) Compare with $\sum 1/n$. Note that

$$\lim_{n \rightarrow \infty} \frac{\sin \frac{1}{n}}{\frac{1}{n}} = \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

And from the last part, the harmonic series diverges. So the series

$$\sum_{n=3}^{\infty} \sin \frac{1}{n}$$

diverges.

3. Justify whether the following series converges absolutely or conditionally, or diverges.

(a) (3 points)

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}.$$

(b) (3 points)

$$\sum_{n=1}^{\infty} \frac{2^n}{n^3}.$$

(c) (4 points)

$$\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}.$$

Solution:

(a) Since

$$\lim_{n \rightarrow \infty} \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} = \lim_{n \rightarrow \infty} \frac{(n+1)n^n}{(n+1)(n+1)^n} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \frac{1}{e} < 1.$$

By Ratio Test, the series converges absolutely.

(b) Note that

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \frac{2}{(\sqrt[n]{n})^3} = 2 > 1.$$

By Root Test, the series diverges.

(c) Note that the sequence $\{1/n\}$ is positive and decreasing on $[1, \infty]$ and $1/n \rightarrow 0$ as $n \rightarrow \infty$. By the alternating series test, it converges. Further, since $\sum 1/n$ diverges. The series converges conditionally.